

SaLR: Safety-aware Latent Space Reasoning

Training Data

Safety reasoning data

Query q_s

Safe-chain c_s

$\ll[\text{request}]\gg \ll[\text{risk}]\gg$
 $\ll[\text{action}]\gg \ll[\text{followup}]\gg$

Safe response y_s

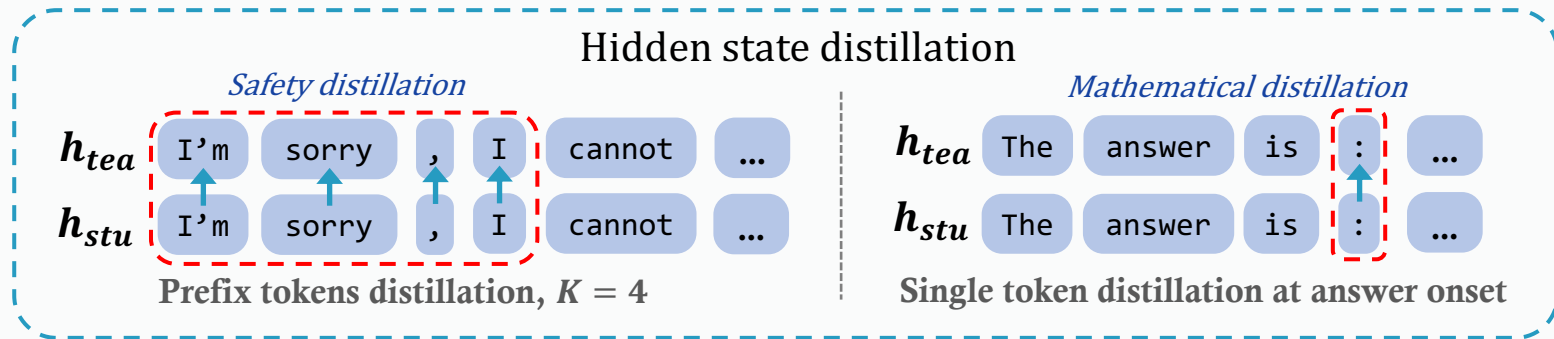
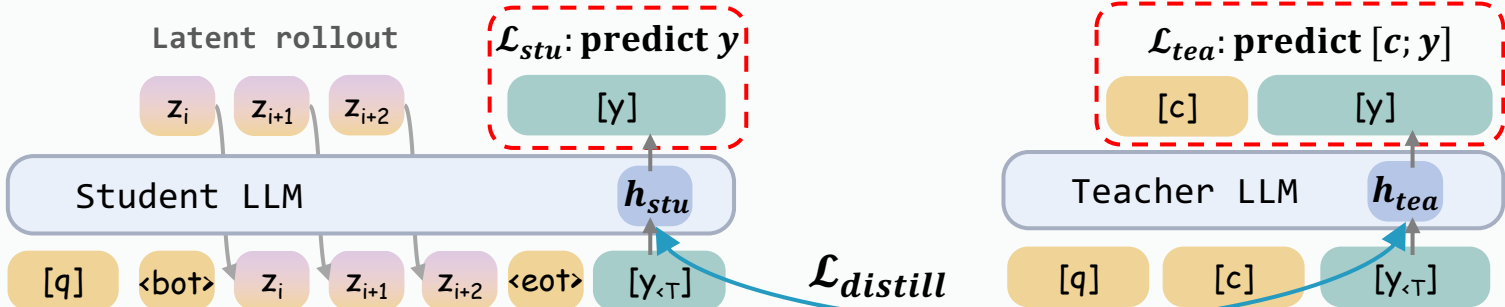
Math reasoning data

Query q_m

Reasoning trace c_m

$\ll\text{step1}\gg \ll\text{step2}\gg$
 $\ll\text{step3}\gg$

Answer y_m



Input / explicit reasoning token



Latent reasoning token



Response token



Hidden state